Account Management KPI BI

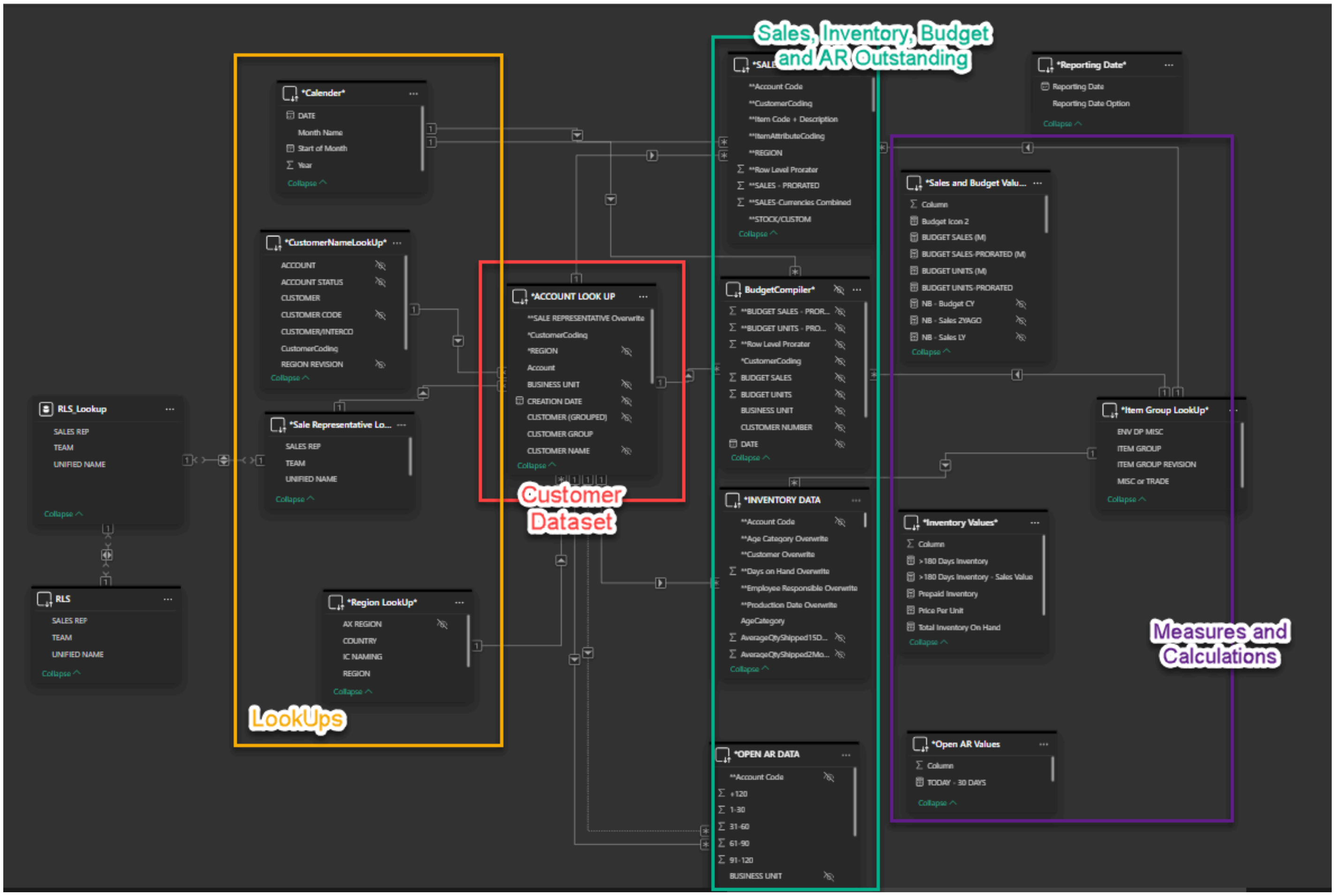
DISCLAIMER
This project was done for an actual client. As such, all confidential items have been blurred out. In addition, all figures have been modified to generate fictitious totals.

INTRO

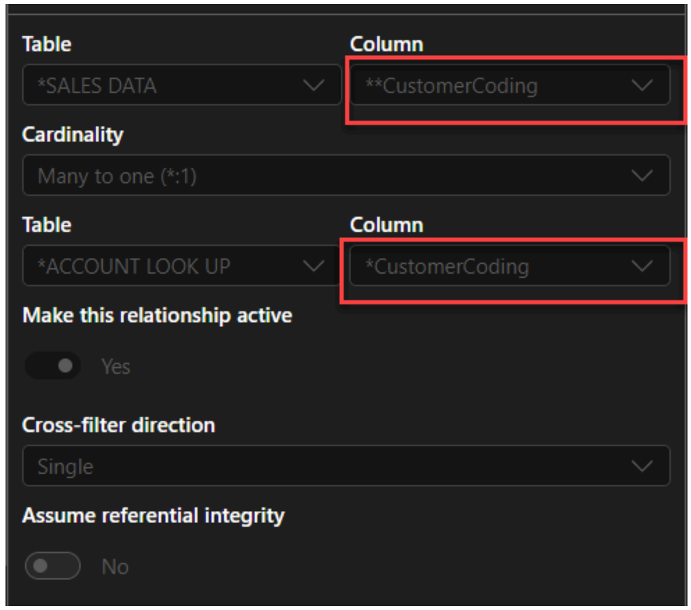
Our sales team **needed a one-stop shop for information about their accounts and territory**. The information required was **detailed sales**(including Y.O.Y and budget variances), **inventory and account receivable outstanding**. **Originally,** it was **created in Microsoft Excel**. Unfortunately, **it would take more than a week to compile and distribute, delaying monthly insights**. **Using datasets that the IT department created**, which were directly linked to our ERP system, I imported and combined them with external lookup tables. With the new dataset, I **created the Account Management KPI BI**; a tool, created in Microsoft Power BI. This BI contained more functionalities than the excel version, and refreshed daily provided up-to-date information

STEPS to Creating the BI

Setting Up Schema



- I adopted a **Snowflake Schema**. To start, I imported the *Customer Dataset*. It contained Account Codes, Account Name, Location (Region) and current Account Owner. This would serve as the center of the schema where all fact tables would be mapped to.
- I imported the *Sales, Inventory and AR Outstanding* Dataset via **direct query**
- The Customer Dataset did not have a distinct set of account codes as there could be singular accounts in multiple locations. So I created a custom column merging account code and location in the Customer Dataset and all other datasets. This created a unique primary key which linked all tables accordingly



- I created an external *LookUps* excel file. When mapped to the associating dataset, it was used to convert plant locations, account owner names, item types, etc into easy to read naming conventions
- I created static excel files containing our annual budget by account and linked that to the Customer Dataset

Creating Measures and Custom Columns

- <u>Pro-rating for Historical and Budget Values</u>
    - In order to have YTD Sale vs Last Year and YTD Budget variances, I need to create measures that would reflect pro-rated portions of historical sales and budgets. This ensured that performance was compared to average targets ie, 3 days of 2025 vs **average** daily sales; 2024 x 3
    - To do this, I created a pro-rating column for the Sales and Budget datasets which would determine how much of their value is incorporated in Y.O.Y and YTD comparisons
    - Rule was:
        - If the month has passed, fully include associating historical sales and budget.
        - If the month has not passed, then exclude historical sales and budget.
        - If the month matches the current month, then totals to include = Working days past in that month/ total working days of that month * totals

```
**Row Level Prorater =
VAR CurrentYear = YEAR(TODAY())
VAR CurrentMonth = MONTH(TODAY())
VAR CurrentDay = DAY(TODAY())
VAR RowYear = YEAR('SALES DATA'[CALENDAR].[Date])
VAR RowMonth = MONTH('SALES DATA'[CALENDAR].[Date])

-- Calculate total weekdays in the current month
VAR StartOfMonth1 = DATE(YEAR(TODAY()-1), MONTH(TODAY()), 1)
VAR EndOfMonth1 = EOMONTH(TODAY()-1, 0)
VAR TotalWeekdaysInMonth =
    SUMX(
        ADDCOLUMNS(
            CALENDAR(StartOfMonth1, EndOfMonth1),
            "IsWeekday", IF(WEEKDAY([Date], 2) <= 5, 1, 0)
        ), [IsWeekday]
    )

-- Calculate weekdays passed in the current month
VAR WeekdaysPassed =
    SUMX(
        ADDCOLUMNS(
            CALENDAR(StartOfMonth1, TODAY()-1),
            "IsWeekday", IF(WEEKDAY([Date], 2) <= 5, 1, 0)
        ), [IsWeekday]
    )

RETURN
    IF(
        RowYear = CurrentYear, 1,  -- If the year is the current year, rate = 1
        IF(
            RowMonth < CurrentMonth, 1,  -- If the month is in the past, rate = 1
            IF(
                RowMonth = CurrentMonth,
                DIVIDE(WeekdaysPassed, TotalWeekdaysInMonth),  -- Weekdays passed / Total weekdays
                0  -- Future months = 0
            )
        )
```

- <u>Sales Values</u>

  Some of the measures created were as follows. *Reporting Date* is set to yesterday, or the last working day.

  - Current Year's YTD. Also for units and price

    ```
    Pro. Sales CY = calculate([SALES-PRORATED
    (M)],'Calender*'[Year]=year(max('Reporting Date*'[Reporting Date])))
    ```

  - Last Year's YTD using the proration mentioned above

    ```
    Pro. Sales LY = calculate([SALES-PRORATED
    (M)],'Calender*'[Year]=year(max('Reporting Date*'[Reporting Date]))-1)
    ```

  - Variances

    ```
    Pro. Sales YTD Var = [Pro. Sales CY]-[Pro. Sales LY]
    Pro. Sales YTD Var (%) = iferror([Pro. Sales YTD Var]/[Pro. Sales LY],1)
    ```

  - Calendar Sales Measures. This was used for the Historical Sales tab which allows users to see sales data from 2022 and after.

    ```
    Total Sales Last Year (H) = CALCULATE(
    ```

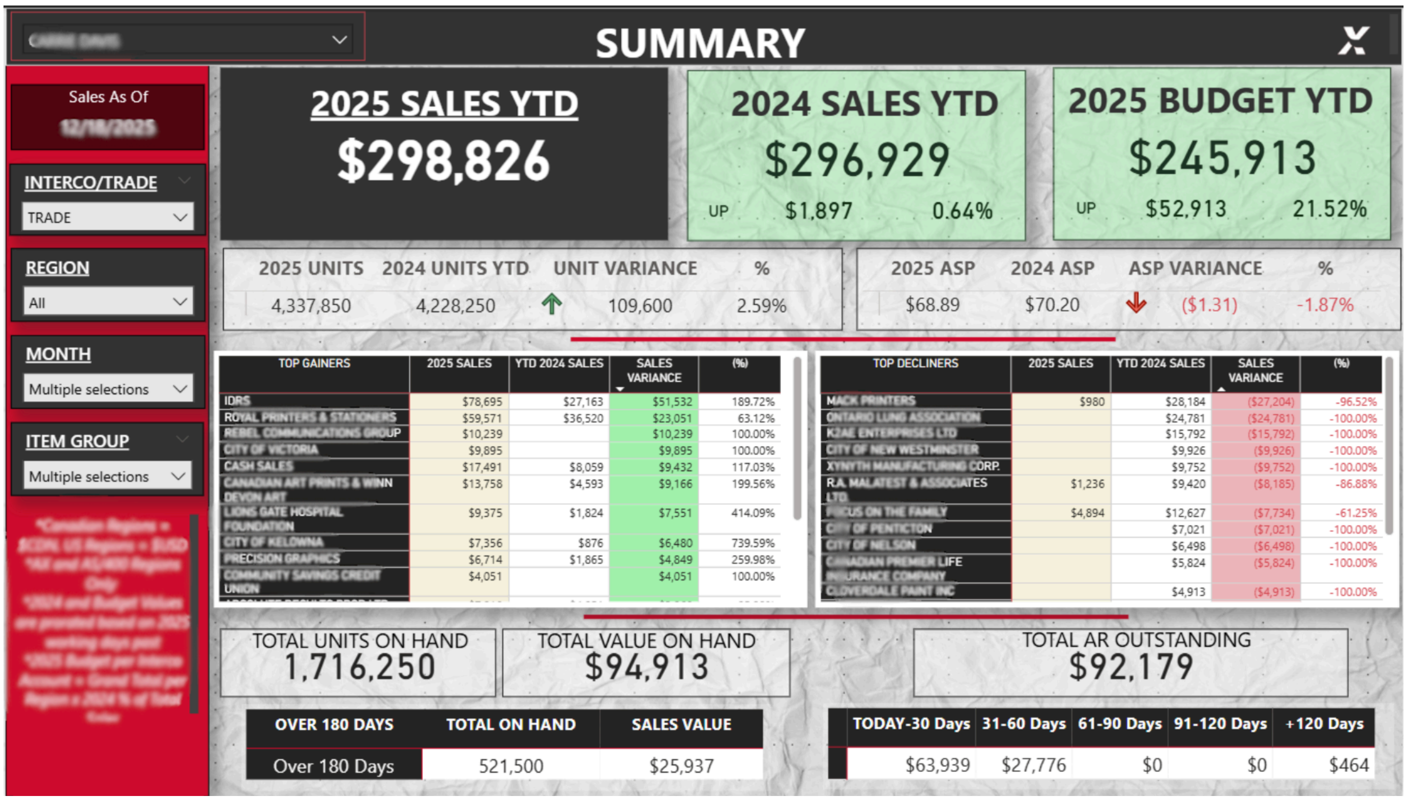
```
    'Sales and Budget Values*'[SALES
(M)],DATEADD('Calender*'[DATE].[Date],-1,YEAR))
```

- Inventory Values
    - The dataset provided by IT already came with aging brackets. Still, I needed to create additional ones for the summary tab. Such as Inventory older than 180 days
    - ```
      >180 Days Inventory = CALCULATE([Total Inventory On Hand],'INVENTORY
      DATA'[dayininventory]>180)
      ```
- AR Values
    - Similar to Inventory, IT did a great job of including the aging groups into the dataset. Therefore, there wasn't many measures to create aside from summing measures
      ```

Creating the Dashboards and Tabs
Typically, Power BI Dashboard layout contains rectangular visuals snapped to a grid. However, after years of knowing what the users would need, I aimed for displaying as much information without creating excessive clutter. In addition, I wanted to emphasize utilizing appealing aesthetics by incorporating the company's branding colours and making the end result "look pretty".

<u>Summary Dashboard</u>

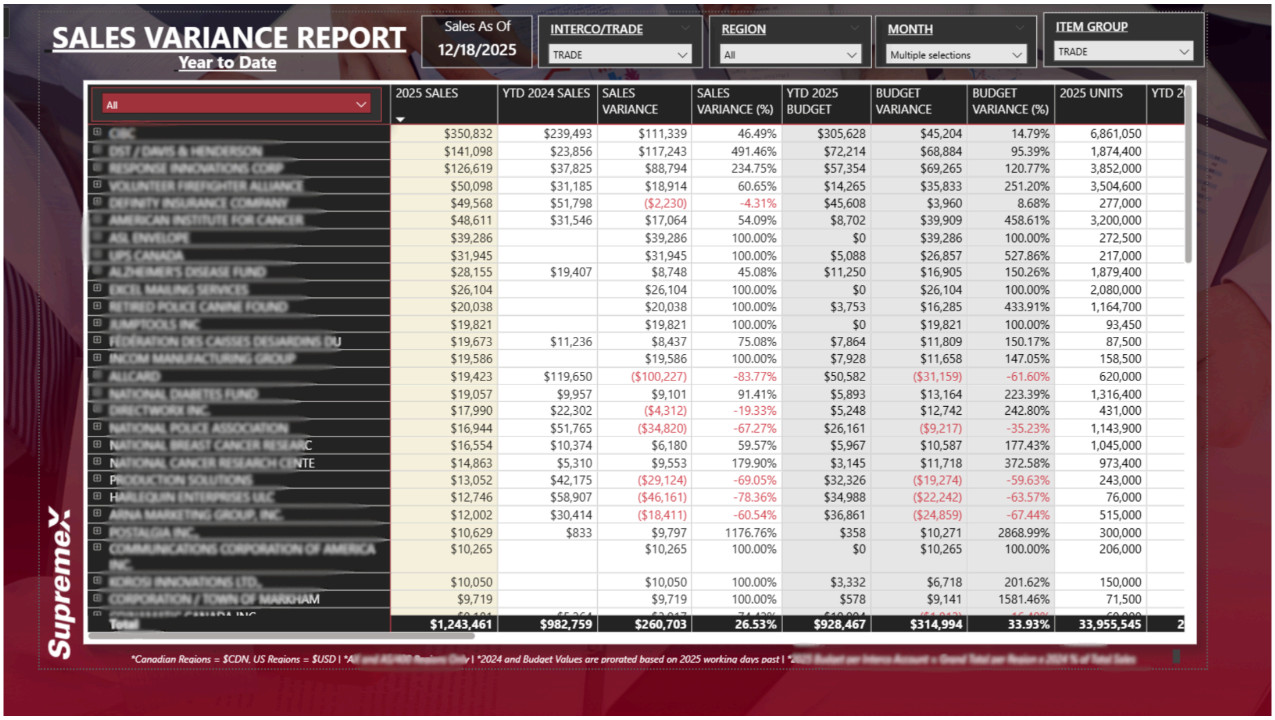
- The Dashboard acts as the one-stop shop feature of the BI. It includes YTD figures as well as variances to last year. The emphasis was less about visuals (bar and line charts) and more about plainly displaying KPIs
- The below show what a sales representative would see upon opening the Summary tab



<u>Sales, Inventory and AR Outstanding Tabs</u>

- As these are meant to provide detailed information, the goal was to create filterable and exportable "datadumps". To add to aesthetics, I imported images that I created in **Canva**, as the tab's wallpaper.

Details Sales Tab



SALES VARIANCE REPORT
Year to Date

Sales As Of 12/18/2025 | INTERCO/TRADE: TRADE | REGION: All | MONTH: Multiple selections | ITEM GROUP: TRADE

All

	2025 SALES	YTD 2024 SALES	SALES VARIANCE	SALES VARIANCE (%)	YTD 2025 BUDGET	BUDGET VARIANCE	BUDGET VARIANCE (%)	2025 UNITS	YTD 2
[illegible]	$350,832	$239,493	$111,339	46.49%	$305,628	$45,204	14.79%	6,861,050	
[illegible]	$141,098	$23,856	$117,243	491.46%	$72,214	$68,884	95.39%	1,874,400	
[illegible]	$126,619	$37,825	$88,794	234.75%	$57,354	$69,265	120.77%	3,852,000	
[illegible]	$50,098	$31,185	$18,914	60.65%	$14,265	$35,833	251.20%	3,504,600	
[illegible]	$49,568	$51,798	($2,230)	-4.31%	$45,608	$3,960	8.68%	277,000	
[illegible]	$48,611	$31,546	$17,064	54.09%	$8,702	$39,909	458.61%	3,200,000	
[illegible]	$39,286		$39,286	100.00%	$0	$39,286	100.00%	272,500	
[illegible]	$31,945		$31,945	100.00%	$5,088	$26,857	527.86%	217,000	
[illegible]	$28,155	$19,407	$8,748	45.08%	$11,250	$16,905	150.26%	1,879,400	
[illegible]	$26,104		$26,104	100.00%	$0	$26,104	100.00%	2,080,000	
[illegible]	$20,038		$20,038	100.00%	$3,753	$16,285	433.91%	1,164,700	
[illegible]	$19,821		$19,821	100.00%	$0	$19,821	100.00%	93,450	
[illegible]	$19,673	$11,236	$8,437	75.08%	$7,864	$11,809	150.17%	87,500	
[illegible]	$19,586		$19,586	100.00%	$7,928	$11,658	147.05%	158,500	
[illegible]	$19,423	$119,650	($100,227)	-83.77%	$50,582	($31,159)	-61.60%	620,000	
[illegible]	$19,057	$9,957	$9,101	91.41%	$5,893	$13,164	223.39%	1,316,400	
[illegible]	$17,990	$22,302	($4,312)	-19.33%	$5,248	$12,742	242.80%	431,000	
[illegible]	$16,944	$51,765	($34,820)	-67.27%	$26,161	($9,217)	-35.23%	1,143,900	
[illegible]	$16,554	$10,374	$6,180	59.57%	$5,967	$10,587	177.43%	1,045,000	
[illegible]	$14,863	$5,310	$9,553	179.90%	$3,145	$11,718	372.58%	973,400	
[illegible]	$13,052	$42,175	($29,124)	-69.05%	$32,326	($19,274)	-59.63%	243,000	
[illegible]	$12,746	$58,907	($46,161)	-78.36%	$34,988	($22,242)	-63.57%	76,000	
[illegible]	$12,002	$30,414	($18,411)	-60.54%	$36,861	($24,859)	-67.44%	515,000	
[illegible]	$10,629	$833	$9,797	1176.76%	$358	$10,271	2868.99%	300,000	
[illegible]	$10,265		$10,265	100.00%	$0	$10,265	100.00%	206,000	
[illegible]	$10,050		$10,050	100.00%	$3,332	$6,718	201.62%	150,000	
[illegible]	$9,719		$9,719	100.00%	$578	$9,141	1581.46%	71,500	
Total	$1,243,461	$982,759	$260,703	26.53%	$928,467	$314,994	33.93%	33,955,545	2

*Canadian Regions = $CDN, US Regions = $USD | *[illegible] | *2024 and Budget Values are prorated based on 2025 working days past | *[illegible]

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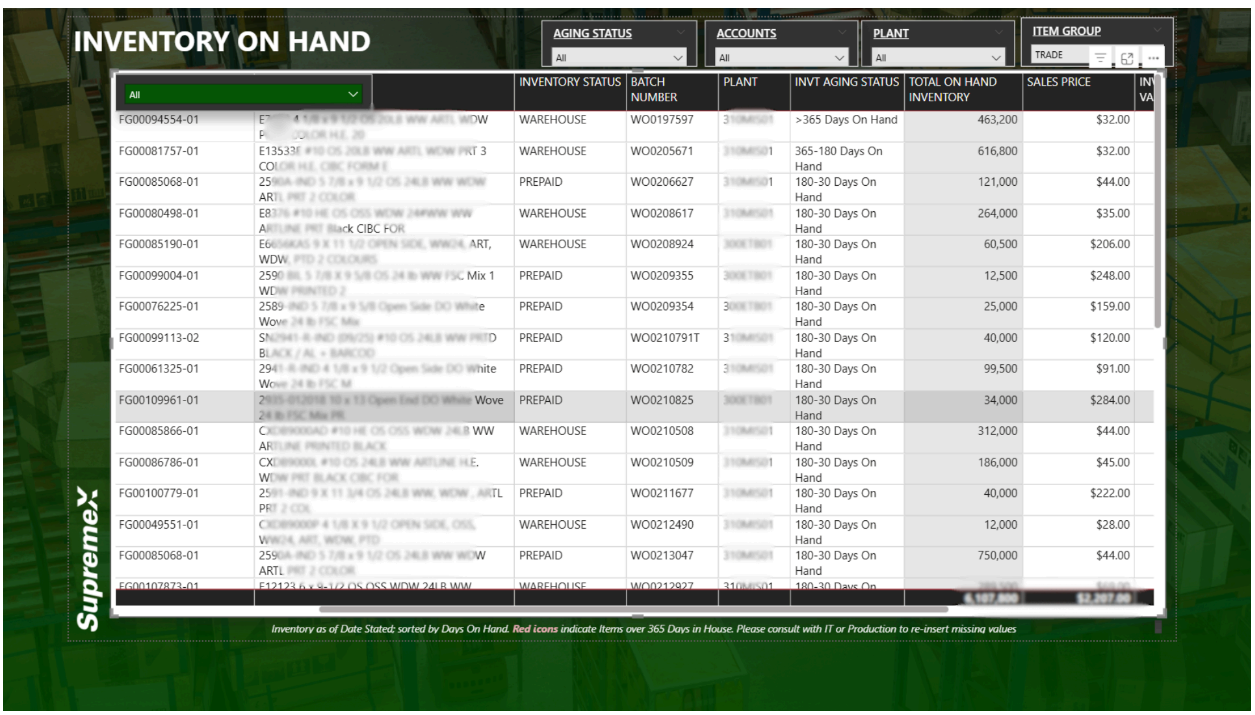
Inventory Tab



INVENTORY ON HAND

AGING STATUS: All | ACCOUNTS: All | PLANT: All | ITEM GROUP: TRADE

All

		INVENTORY STATUS	BATCH NUMBER	PLANT	INVT AGING STATUS	TOTAL ON HAND INVENTORY	SALES PRICE	IN VA
FG00094554-01	[illegible]	WAREHOUSE	WO0197597	[illegible]	>365 Days On Hand	463,200	$32.00	
FG00081757-01	E13533[illegible]	WAREHOUSE	WO0205671	[illegible]	365-180 Days On Hand	616,800	$32.00	
FG00085068-01	259[illegible]	PREPAID	WO0206627	[illegible]	180-30 Days On Hand	121,000	$44.00	
FG00080498-01	E8[illegible] Black CIBC FOR	WAREHOUSE	WO0208617	[illegible]	180-30 Days On Hand	264,000	$35.00	
FG00085190-01	E6[illegible] ART, WDW[illegible]	WAREHOUSE	WO0208924	[illegible]	180-30 Days On Hand	60,500	$206.00	
FG00099004-01	2590[illegible] Mix 1 WDW[illegible]	PREPAID	WO0209355	[illegible]	180-30 Days On Hand	12,500	$248.00	
FG00076225-01	2589[illegible] White Wove[illegible]	PREPAID	WO0209354	[illegible]	180-30 Days On Hand	25,000	$159.00	
FG00099113-02	SN[illegible]	PREPAID	WO0210791T	[illegible]	180-30 Days On Hand	40,000	$120.00	
FG00061325-01	294[illegible] White Wove[illegible]	PREPAID	WO0210782	[illegible]	180-30 Days On Hand	99,500	$91.00	
FG00109961-01	[illegible] Wove	PREPAID	WO0210825	[illegible]	180-30 Days On Hand	34,000	$284.00	
FG00085866-01	CX[illegible] WW AR[illegible]	WAREHOUSE	WO0210508	[illegible]	180-30 Days On Hand	312,000	$44.00	
FG00086786-01	CX[illegible] WDW[illegible]	WAREHOUSE	WO0210509	[illegible]	180-30 Days On Hand	186,000	$45.00	
FG00100779-01	25[illegible] PR[illegible]	PREPAID	WO0211677	[illegible]	180-30 Days On Hand	40,000	$222.00	
FG00049551-01	[illegible]	WAREHOUSE	WO0212490	[illegible]	180-30 Days On Hand	12,000	$28.00	
FG00085068-01	259[illegible] ARTL[illegible]	PREPAID	WO0213047	[illegible]	180-30 Days On Hand	750,000	$44.00	

Inventory as of Date Stated; sorted by Days On Hand. Red icons indicate Items over 365 Days in House. Please consult with IT or Production to re-insert missing values

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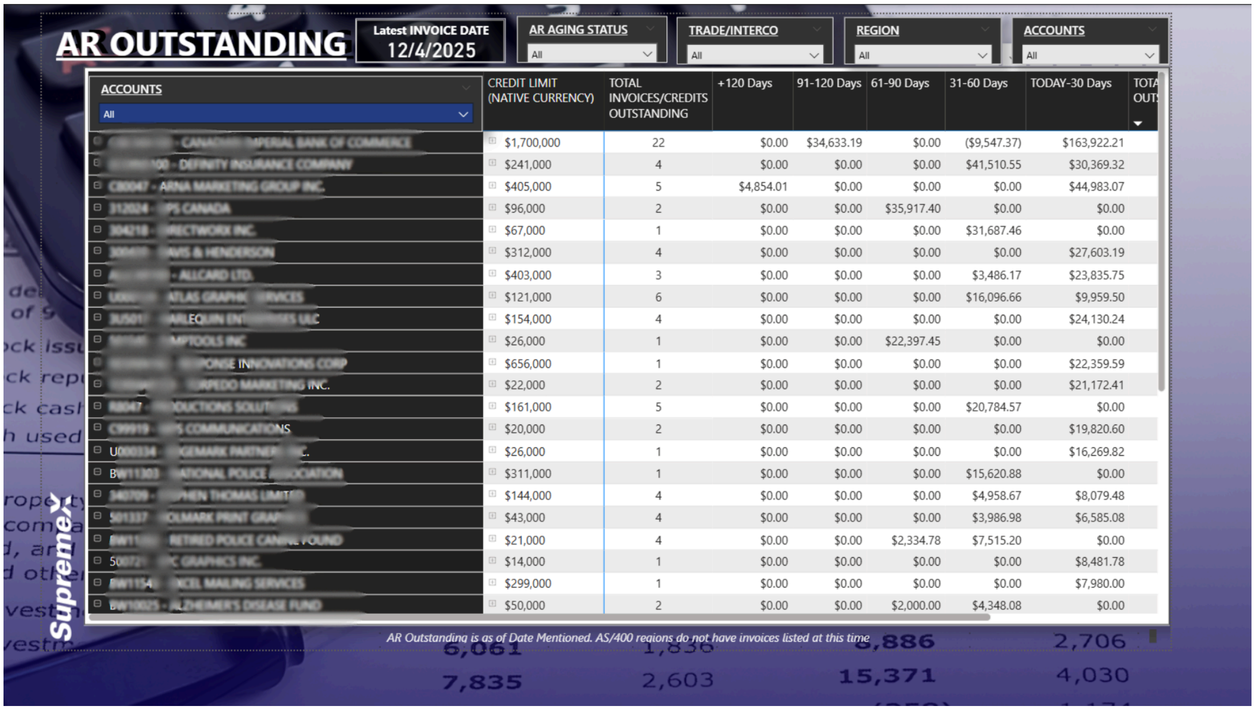
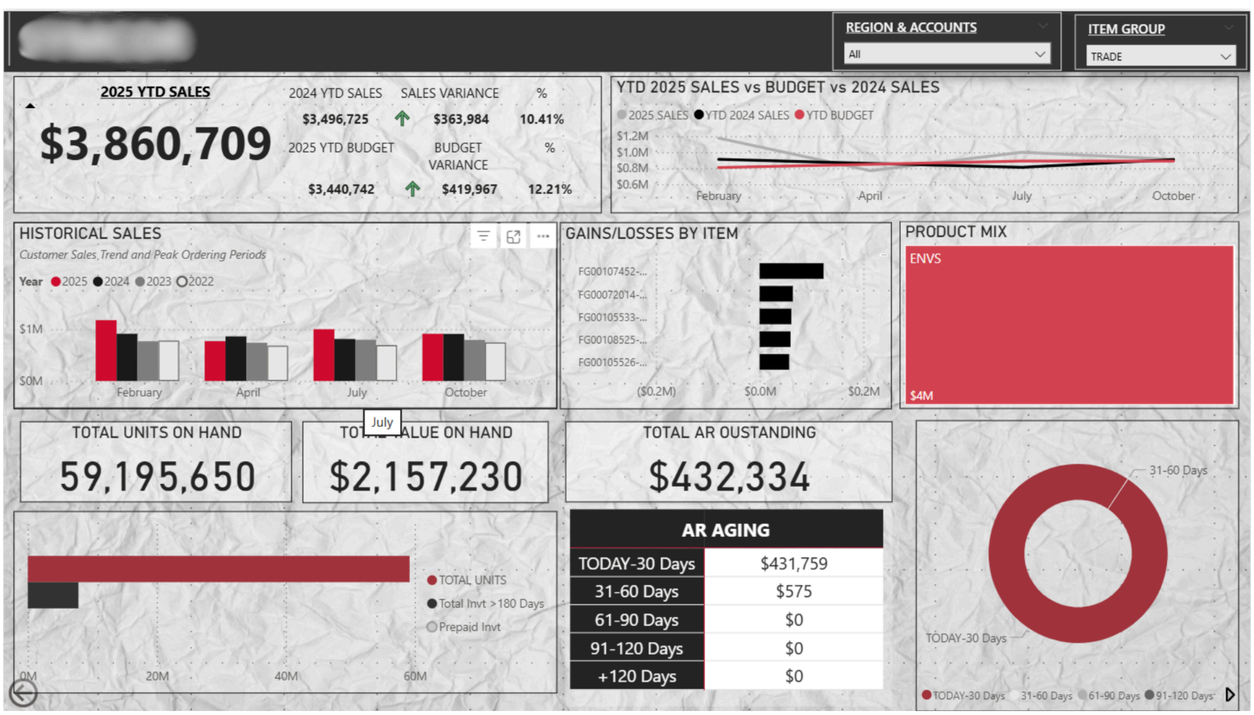
AR Outstanding Tab



<u>Customer Profile Tab</u>

- I wanted to create a tab where users could drill down on any customer, from any of the tabs mentioned above, and get information about that customer. Here, I utilized interactive visuals which contained many tooltips for additional information. The goals was that a user could export this page and use it in their business reviews



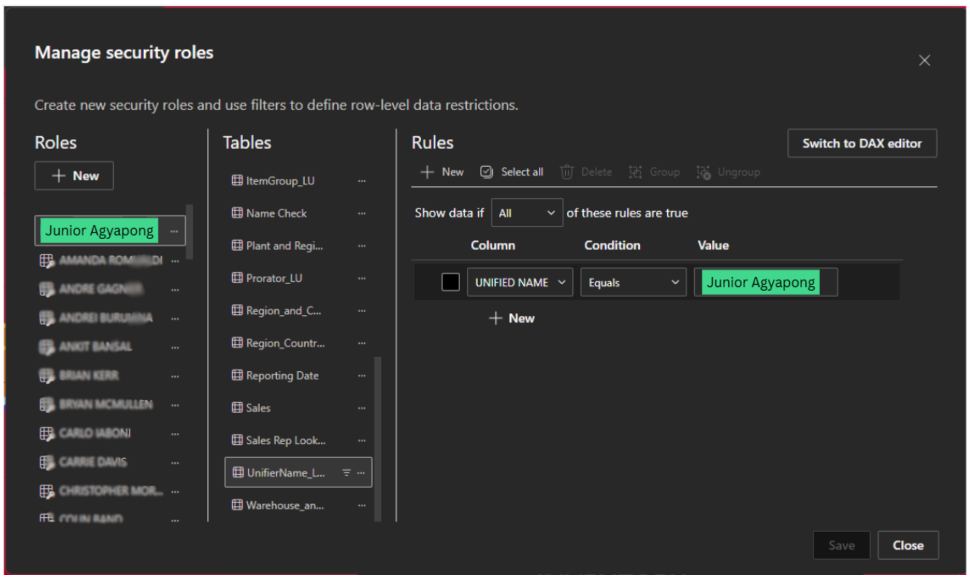
Landing Page

- As there are many tabs, I wanted to create a landing page that would open whenever a user opened their BI. I wanted it to mention their name to add a unique sense of personalization. I also left space so that announcements about issues or updates could be made and immediately viewed.

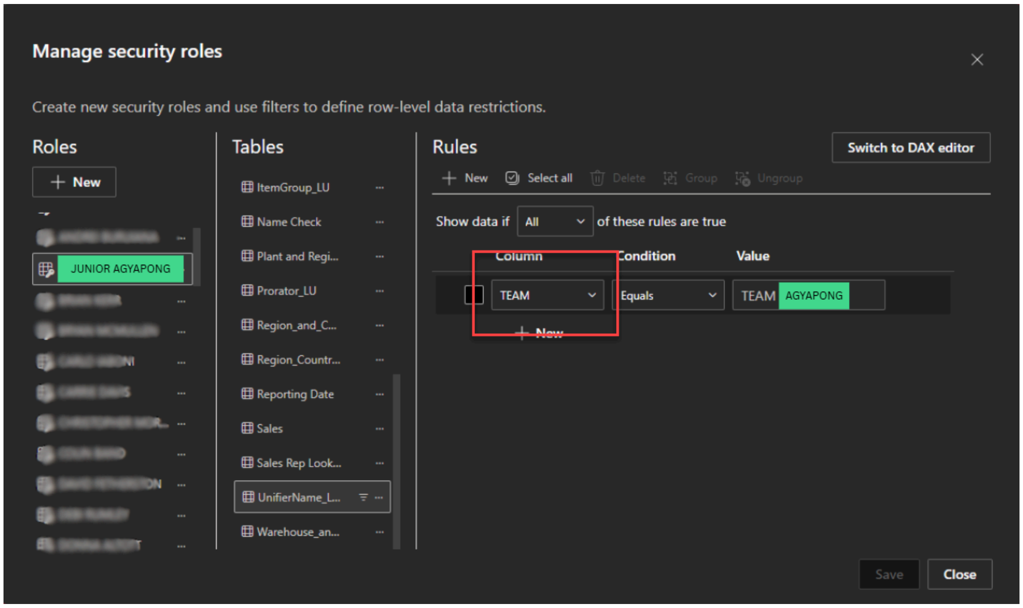
Managing Permissions and Row Level Security

I needed to ensure that account owners only see accounts pertaining to their territory. In addition, I needed to ensure that Sales Managers could see their team's performance and territory.
To do this, I used Row Level Security (RLS). In the LookUp dataset mentioned above, I imported *Account Owner's names and Teams.* There, I set up each RLS so the information that is displayed is prefiltered to the account owner's name. For management, I set up the RLS to display prefiltered information based on the Team's Name. Each team could have multiple account owners

Example of setting up Row Level Security for an Account Owner



Example of setting up Row Level Security for a Sales Manager



Once published, I assigned the RLS to each account owner's email so that the BI would only display a prefiltered report based on the user logging in.

Last Steps

- Setting up the Daily Refresh schedule on Power BI Online
- Creating a BI that would display new values that need to be added to the LookUps file.
- Deploying the BI and hosting tutorials
- Monitoring the tools functionality and accuracy